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File

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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		z1(z1z2)	
Gear 2:		z2(z1z2)	
		------- Gear 1 ----------	Gear 2 -------
Power (kW)	[P]	234.572	
Speed (1/min)	[n]	7000.0	3070.2
Number of load cycles (in mio.)	[NL]	3990.000	1750.000
Torque (Nm)	[T]	320.0	729.6
Application factor	[KA]	1.00	
Required service life (h)	[H]	9500.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1 ----------	Gear 2 -------
Center distance (mm)	[a]	118.531	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	2.5000	
Transverse module (mm)	[mt]	2.8584	
Normal diametral pitch (1/in)	[Pnd]	10.16000	
Transverse diametral pitch (1/in)	[Ptd]	8.88614	
Normal pressure angle (°)	[αn]	18.0000	
Helix angle at reference circle (°)	[β]	29.0000	
Number of teeth	[z]	25	57
Facewidth (mm)	[b]	20.80	18.80
Hand of gear		left	right
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	25.00	138.10
Shaft diameter (mm)	[ds]	25.00	128.54
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1 ----------	Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

------- Gear 1 ---------- Gear 2 -------

Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1 ----------	Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	ISO-VG 46	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	46.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	6.80
Specific density at 15°C (kg/dm³)	[ρ]	0.878
Oil temperature (°C)	[TS]	70.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.789
Root radius factor	[ρfP*]	0.236
	[ρfPmax*]	0.281
Addendum coefficient	[haP*]	1.352
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	36.787
(mm)	[a0.e/i]	36.557 /36.485
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.815 / 0.824

Reference profile Gear 2

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.509
Root radius factor	[ρfP*]	0.361
	[ρfPmax*]	0.406
Addendum coefficient	[haP*]	1.613
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000

	not topping	
Manufacturing center distance (mm)	[a0]	81.797
(mm)	[a0.e/i]	81.627 /81.537
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.989 / 0.994

4.1.1 Information on final machining

		Gear 1	Gear 2
Dedendum reference profile	[hfP*]	1.789	1.509
Tooth root radius reference profile	[ρfP*]	0.236	0.361
Addendum reference profile	[haP*]	1.352	1.613
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000

Type of profile modification:	for uniform mesh		
Tip relief (µm)	[Ca L/R]	13.5 /13.5	14.7 /14.7
Root relief (µm)	[Cf L/R]	17.0 /17.0	17.0 /17.0

4.2 Basic data

Overall transmission ratio	[itot]	-2.280
Gear ratio	[u]	2.280
Transverse module (mm)	[mt]	2.858
Transverse pressure angle (°)	[αt]	20.380
Working pressure angle (°)	[αwt]	22.054
(°)	[αwt.e/i]	22.075 /22.033
Working pressure angle at normal section (°)	[αwn]	19.462
Base helix angle (°)	[βb]	27.457
Helix angle at operating pitch circle (°)	[βw]	29.276
Reference center distance (mm)	[ad]	117.194
Sum of profile shift coefficients	[Σxi]	0.5560

		Gear 1	Gear 2
Profile shift coefficient	[x]	0.4230	0.1330
Generating profile shift coefficient	[xE.e/i]	0.3799/ 0.3553	0.0746/ 0.0438
Virtual gear no. of teeth	[zn]	36.301	82.767
Involute length (mm)	[l_dFa-l_dFf]	7.122	7.359

4.3 Diameters and their allowances

		Gear 1	Gear 2
Reference diameter (mm)	[d]	71.460	162.928
Base diameter (mm)	[db]	66.987	152.729
Tip alteration (mm)	[k*mn]	0.000	0.000
Tip diameter (mm)	[da]	80.334	171.660
(mm)	[da.e/i]	80.334 /80.299	171.660 /171.620
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.035	0.000 / -0.040
Tip form diameter (mm)	[dFa]	80.334	171.660
(mm)	[dFa.e/i]	80.334 /80.299	171.660 /171.620
Active tip diameter (mm)	[dNa]	80.334	171.660
(mm)	[dNa.e/i]	80.334 /80.299	171.660 /171.620
V-Circle diameter (mm)	[dv]	73.575	163.593
(mm)	[dv.e/i]	73.359 /73.236	163.301 /163.147

Operating pitch diameter (mm)	[dw]	72.275	164.787
(mm)	[dw.e/i]	72.285 /72.264	164.811 /164.762
Root diameter (mm)	[df]	64.632	156.047
(mm)	[df.e/i]	64.416 /64.293	155.755 /155.601
Active root diameter (mm)	[dNf]	67.828	159.127
(mm)	[dNf.e/i]	67.856 /67.813	159.171 /159.101
Root form diameter (mm)	[dFf]	67.419	158.024
(mm)	[dFf.e/i]	67.351 /67.316	157.810 /157.700

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.340	0.385
Effective tip clearance (mm)	[c.e/i]	0.598 / 0.469	0.592 / 0.475
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.270 / 0.231	0.736 / 0.645
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	4.437	4.366
(mm)	[ha.e/i]	4.437 / 4.420	4.366 / 4.346
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	3.414	3.440
(mm)	[hf.e/i]	3.522 / 3.583	3.587 / 3.664
Tooth height (mm)	[h]	7.851	7.806

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	37.929 /37.874	29.398 /29.365
Roll angle to dNf (°)	[ξNf.e/i]	9.262 / 9.028	16.816 /16.722
Roll angle at dFf (°)	[ξFf.e/i]	5.988 / 5.686	14.902 /14.736

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8457	1.6573
Normal tooth thickness at tip circle (mm)	[san]	0.845	0.867
(mm)	[san.e/i]	0.788 / 0.725	0.786 / 0.717
Normal tooth thickness at tip form circle (mm)	[sFan]	0.845	0.867
(mm)	[sFan.e/i]	0.788 / 0.725	0.786 / 0.717
Normal space width at root circle (mm)	[efn]	0.000	1.823
(mm)	[efn.e/i]	0.000 / 0.000	1.860 / 1.880
Pitch on reference circle (mm)	[pt]	8.980	8.980
Base pitch (mm)	[pbt]	8.418	8.418
Transverse pitch on contact-path (mm)	[pet]	8.418	8.418
Lead height (mm)	[pz]	405.003	923.408
Axial pitch (mm)	[px]	16.200	16.200

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	9.072	8.694
Specific sliding at the tip	[ζa]	0.558	0.690

Specific sliding at the root	[ζf]	-2.228	-1.263
Mean specific sliding	[ζm]	0.623	
Sliding factor on tip	[Kga]	0.342	0.328
Sliding factor on root	[Kgf]	-0.328	-0.342

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	42.400
Transverse contact ratio	[εα]	2.001
	[εα.e/m/i]	2.007 / 1.997/ 1.987
Overlap ratio	[εβ]	1.160
Total contact ratio	[εγ]	3.162
	[εγ.e/m/i]	3.167 / 3.157/ 3.147
Length of path of contact (mm)	[gα]	16.848
(mm)	[gα.e/m/i]	16.894 /16.810/16.726
Approach path of contact (mm)	[gf]	8.245
(mm)	[gf.e/m/i]	8.277 / 8.223/ 8.169
Recess path of contact (mm)	[ga]	8.603
(mm)	[ga.e/m/i]	8.617 / 8.587/ 8.557

		Gear 1	Gear 2
Addendum contact ratio	[ε]	1.022	0.979
	[ε.e/i]	1.024 / 1.017	0.983 / 0.970
Length T1-A and T2-A (mm)	[T1A,T2A]	5.324	39.182
(mm)	[.e/i]	5.277 / 5.414	39.182 / 39.138
Length T1-B and T2-B (mm)	[T1B,T2B]	5.336	39.169
(mm)	[.e/i]	5.336 / 13.722	39.123 / 30.830
Length T1-C and T2-C (mm)	[T1C,T2C]	13.569	30.937
(mm)	[.e/i]	13.555 / 13.583	30.904 / 30.969
Length T1-D and T2-D (mm)	[T1D,T2D]	22.160	22.346
(mm)	[.e/i]	22.113 / 13.832	22.346 / 30.720
Length T1-E and T2-E (mm)	[T1E,T2E]	22.172	22.334
(mm)	[.e/i]	22.172 / 22.140	22.287 / 22.412
Length T1-T2 (mm)	[T1T2]	44.506	
(mm)	[.e/i]	44.459 / 44.552	
Diameter of single contact point B (mm)	[d-B]	67.831	171.648
(mm)	[d-B.e/i]	67.831 / 72.391	171.606 /164.706
Diameter of single contact point D (mm)	[d-D]	80.321	159.134
(mm)	[d-D.e/i]	80.269 / 72.474	159.134 /164.624

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	8956.11	
Axial force (N)	[Fa]	4964.45	4964.45
Radial force (N)	[Fr]	3327.18	3327.18
Normal force (N)	[Fnorm]	10766.97	10766.97
Nominal circumferential force per mm (N/mm)	[w]	476.39	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	8855.09	

Axial force (N)	[Faw]	4964.45	4964.45
Radial force (N)	[Frw]	3587.38	
Circumferential speed reference circle (m/s)	[v]	26.19	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	26.49	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.6
Running-in value (µm)	[yf]	0.7
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	0.745
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	10.231
Theoretical single stiffness (N/mm/µm)	[c'th]	19.639
Contact stiffness (N/mm/µm)	[cγα]	17.915
Contact stiffness (N/mm/µm)	[cγβ]	15.228
Reduced mass (kg/mm)	[mRed]	0.01336
Resonance speed (min-1)	[nE1]	13990
Resonance ratio (-)	[N]	0.500
Subcritical range		

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.3

5.4 K factors

Dynamic factor	[Kv]	1.073
Factor $K_{H\beta}$, own input		
Face load factors		
- Flank	[KHβ]	1.287
- Tooth root	[KFβ]	1.191
- Scuffing	[KBβ]	1.287
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.000
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.3553	0.0438
Tooth form factor	[YF]	0.710	0.800
Stress correction factor	[YS]	2.199	2.075
Load application angle (°)	[αFen]	14.281	17.248
Load distribution influence factor	[fε]	0.627	
Load application diameter, virtual spur gear (mm)	[d_{en}]	90.355	207.298
Load application diameter (mm)	[d_{en}]	71.061	163.308
Bending moment arm (mm)	[hFe]	2.421	2.767
(-)	[hFe*]	0.969	1.107
Tooth root thickness at critical cross section (mm)	[sFn]	5.716	5.715
(-)	[sFn*]	2.287	2.286

Root fillet radius at critical cross section (mm)	[ρF]	1.252	1.281
(-)	[ρF*]	0.501	0.512
Tangent contact point, virtual spur gear (mm)	[x,y]	2.858 /32.464	
Tangent contact point, virtual spur gear (mm)	[x,y]		2.857 /78.267
Diameter at critical cross section (mm)	[d_{sFn}]	65.179	156.638
Tangent at critical cross section (°)	[α_{sFn}]	30.000	30.000
Notch parameter	[q_s]	2.284	2.231
Helix angle factor	[Yβ]	1.133	
Deep tooth factor	[YDT]	1.000	
Gear rim thickness (mm)	[sr]	19.65	8.75
Gear rim factor	[YB]	1.00	1.11
Effective facewidth (mm)	[beff]	20.80	18.80
Nominal stress at tooth root (N/mm²)	[σF0]	304.99	397.73
Tooth root stress (N/mm²)	[σF]	389.87	508.43
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.998	0.997
Surface factor	[YRrelT]	0.957	0.957
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	0.866	0.880
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.827	0.840
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	501.87	509.97
Limit strength tooth root (N/mm²)	[σFG]	702.62	713.95

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	1.80	1.40
Transmittable power (kW)	[kWRating]	301.96	235.28

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.233	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.707	
Helix angle factor	[Zβ]	1.069	
Effective facewidth (mm)	[beff]	18.80	
Nominal contact stress (N/mm²)	[σH0]	991.98	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1166.02	
Coefficient [fZCa] 1.07 (Helical gear sets with suitable flank modifications acc. to manufacturers experience)			
Single tooth contact factor	[ZB,ZD]	1.03	1.03
Contact stress (N/mm²)	[σHB, σHD]	1206.14	1206.14
Lubricant coefficient for N_L	[ZL]	0.931	0.931
Speed factor at N_L	[ZV]	1.028	1.028
Roughness factor for N_L	[ZR]	0.962	0.962
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.874	0.897
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.805	0.826
Limited pitting is permissible: No			

Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1207.84	1238.76
Pitting stress limit (N/mm²)	[σHG]	1207.84	1238.76

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.04	1.06
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.00	1.03
Safety regarding transmittable torque	[SHBD²]	1.00	1.05
Transmittable power (kW)	[kWRating]	235.23	247.43

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Calculation has not been carried out, lubricant: Load stage micropitting test not known

9 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
with C_{eq} according to	ISO/TS 6336-20, Annex B		
Optimal tip relief (µm)	[Ceff]	26.59	
C_a taken as optimal in the calculation. 0=no, 1=yes		1	1

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.292	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	12
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	18.800	
Applicable circumferential force/facewidth (N/mm)	[wBt]	658.212	
$K_{B\gamma}$ = 1.292, $w_{Bt} \cdot K_{B\gamma}$ = 850.596			
Angle factor	[Xαβ]	0.978	
ε_1 =1.022, ε_2 =0.979			

9.1 Flash temperature-criteria

		---------- Pair ----------	
Lubricant factor	[XL]	0.881	
Tooth mass temperature (°C)	[θMi]	101.18	
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$			
Average flash temperature (Δ°C)	[θflm]	66.34	
Scuffing temperature (°C)	[θS]	365.60	
Γ coordinates (point of highest temperature)	[Γ]	-0.595	
[Γ.A]= -0.608, [Γ.E]= 0.634			
Maximum flash temperature (Δ°C)	[θflamax]	149.38	
Maximum contact temperature (°C)	[θBmax]	250.56	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Approach factor	[XJ]	1.000	
Load sharing factor	[XΓ]	0.720	
Dynamic viscosity (mPa*s)	[ηM]	12.46	(70.0 °C)
Coefficient of friction	[μ_m]	0.053	
Required safety	[SBmin]	2.000	
Margin of safety for scuffing, flash temperature	[SB]	1.637	

9.2 Integral temperature-criteria

		Pair
Lubricant factor	[XL]	1.000
Tooth mass temperature (°C)	[θMC]	85.90
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$		
Integral scuffing temperature (°C)	[θSint]	365.79
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058
Running-in factor, well run in	[XE]	1.000
Contact ratio factor	[Xε]	0.217
Dynamic viscosity (mPa*s)	[ηOil]	12.46 (70.0 °C)
Mean coefficient of friction	[μ_m]	0.056
Geometry factor	[XBE]	0.309
Meshing factor	[XQ]	1.000
Tip relief factor	[XCa]	1.987
Mean flash temperature (Δ°C)	[θflaint]	22.71
Integral tooth flank temperature (°C)	[θint]	119.97
Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	3.049
Safety factor for transmitted moment (int.-T.)	[SSL]	5.920

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		Gear 1	Gear 2
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.070 / -0.110	-0.095 / -0.145

10.2 Base tangent lengths

		Gear 1	Gear 2
Number of teeth spanned	[k]	5.000	9.000
Base tangent length (no backlash) (mm)	[Wk]	35.206	65.838
Base tangent length with allowance (mm)	[Wk.e/i]	35.139 /35.101	65.748 /65.701
(mm)	[ΔWk.e/i]	-0.067 / -0.105	-0.090 / -0.138
Diameter of measuring circle (mm)	[dMWk.m]	73.881	163.486
> Gear 2: Base tangent length cannot be measured (gear too thin)			
Point of contact in profile-modified area (0=no, 1=yes)		0	1
The measurement is inaccurate!			

10.3 Measurement over balls and pins

		Gear 1	Gear 2
Theoretical diameter of ball/pin (mm)	[DM]	4.457	4.182
Effective diameter of ball/pin (mm)	[DMeff]	4.250	4.250
Radial single-ball measurement, no backlash (mm)	[MrK]	39.505	84.702
Radial single-ball measurement (mm)	[MrK.e/i]	39.420 /39.371	84.568 /84.497
Point of contact in profile-modified area (0=no, 1=yes)		1	1
The measurement is inaccurate!			
Diameter of measuring circle (mm)	[dMMr.m]	72.966	163.435
Diametral measurement over two balls, no backlash (mm)	[MdK]	78.862	169.342
Diametral measurement over two balls (mm)	[MdK.e/i]	78.693 /78.595	169.073 /168.931
Diametral measurement over pins, no backlash (mm)	[MdR]	79.010	169.405
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	78.840 /78.742	169.136 /168.993
Measurement over 2 pins, free, AGMA 2002 (mm)	[dk2f.e/i]	78.619 /78.522	169.045 /168.902
Measurement over 2 pins, transverse, AGMA 2002 (mm)	[dk2t.e/i]	78.984 /78.886	169.198 /169.056
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	78.840 /78.742	169.136 /168.993

10.4 Tooth thickness

		------- Gear 1	---------- Gear 2 -------
Medium tip diameter (mm)	[da.m]	80.317	171.640
Reference chordal height from da.m (mm)	[hac]	4.485	4.376
Tooth thickness at height hac, chord (mm)	[sc]	4.612	4.143
(mm)	[sc.e/i]	4.543 / 4.504	4.048 / 3.999
Tooth thickness on reference circle, arc (mm)	[sn]	4.614	4.143
(mm)	[sn.e/i]	4.544 / 4.504	4.048 / 3.998

10.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.014 / -0.014
Radial backlash (mm)	[jrw.e/i]	0.385 / 0.219
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.309 / 0.177
Normal backlash (mm)	[jnw.e/i]	0.253 / 0.146
Center distance allowances (mm)	[Aa.e/i]	0.018 / -0.018
Backlash free center distance (mm)	[aControl.e/i]	118.294 / 118.163
Backlash free center distance, allowances (mm)	[jta]	-0.237 / -0.367

		------- Gear 1	---------- Gear 2 -------
dNf.i with aControl (mm)	[dNf0.i]	67.546	158.584
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	0.097	0.387
Tip clearance (mm)	[c0.i(aControl)]	0.119	0.125

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.2149 / 0.1228	(0°12'54" / 0°7'22")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.4900 / 0.2800	(0°29'24" / 0°16'48")

11 Gear tolerances

		------- Gear 1	---------- Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±8.50	±8.50
Base pitch tolerance (µm)	[fpbT]	±7.97	±7.97
Sector pitch tolerance (µm)	[Fpz/8T]	±17.00	±19.00
Cumulative pitch tolerance, total (µm)	[FpT]	26.00	30.00
Adjacent pitch difference tolerance (µm)	[fuT]	12.00	12.00
Profile form tolerance (µm)	[ffαT]	9.00	9.00
Profile slope tolerance (µm)	[fHαT]	±7.00	±7.50
Profile tolerance, total (µm)	[FαT]	12.00	12.00
Helix form tolerance (µm)	[ffβT]	9.50	9.50
Helix slope tolerance (µm)	[fHβT]	±8.50	±8.50
Helix tolerance, total (µm)	[FβT]	13.00	13.00
Runout tolerance (µm)	[FrT]	24.00	27.00
Single flank composite tolerance, total (µm)	[FisT]	35.00	38.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	8.50	8.50
Radial composite tolerance, total (µm)	[Fi"]	31.00	37.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	9.50	9.50

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R39	R40
Radial composite tolerance, total (µm)	[FidT]	29.00	39.00

Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	13.00	16.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	13.00
Distance between bearings for calculation (mm)	[LG]	41.60
Maximum value for axis deviation error (µm)	[fΣβ]	± 14.38
Maximum value for axis inclination error (µm)	[fΣδ]	± 28.77

12 Modifications and determination of the tooth form

12.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, arc-like

  Caa = 12.000 µm LCa = 1.260 ·mn dCa = 77.036 mm

- Shortened profile crowning

  Caf = 17.0 µm F1 = 0.500 F2 = 0.250

  rcrown=4261mm, Cαa = 1.469µm, dCa = 77.088mm, Factor1 =1.23874 (combine with tip relief)

- Flank line crowning

  Cb = 5.000 µm

  bx=20.800mm, rcrown=10816mm

Gear 2

Symmetric (both flanks)

- Tip relief, arc-like

  Caa = 12.000 µm LCa = 1.619 ·mn dCa = 168.119 mm

- Flank line crowning

  Cb = 5.000 µm

  bx=18.800mm, rcrown=8836mm

- Shortened profile crowning

  Caf = 17.0 µm F1 = 0.500 F2 = 0.250

  rcrown=3563mm, Cαa = 2.667µm, dCa = 168.197mm, Factor1 =1.58138 (combine with tip relief)

12.1.1 Tip relief verification

		------- Gear 1	---------- Gear 2 -------
Diameter (mm)	[dcheck]	80.249	171.570
Tip relief left/right (µm)	[Ca L/R]	13.0 /13.0	14.2 /14.2
Diameter (mm)	[dNa.i]	80.299	171.620
Tip relief left/right (µm)	[Ca L/R]	13.3 /13.3	14.5 /14.5

12.2 Data for the tooth form calculation

Data not available.

Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	121.947

13.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	0.592	0.899
Total mass (kg)	[Σm]	1.491	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.0004351	0.00516

System	(da+df)/2...di (kg*m²)	[J]	0.001428

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	0.318
Torsion when subjected to nominal torque (°)	[δcr]	0.058
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	82.235

13.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.051
Wear sliding coefficient by Niemann	[ζw]	1.246
Loss factor	[HV]	0.204
Gear power loss (kW)	[PVZ]	2.430
Meshing efficiency (%)	[ηz]	98.964

14 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	9500.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	9947

		------- Gear 1 ----------	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1.105e+04
Tooth flank service life (h)	[H_{Hatt}]	9947	2.268e+04

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 9500.0 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	85.9756	95.5045	41.8879

Damage relative to the system service life (H_{att}, 9947.2 h)

D_{F1} (%)	D_{F2} (%)	D_{H1} (%)	D_{H2} (%)
0.0000	90.0226	100.0000	43.8596

15 Reliability calculation

15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

15.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)

$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	β	t_0	T	R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	420000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	420000	1.300e+00	3.766e+09	1.794e+10	99.5455
2	Tooth root	184211	1.700e+00	1.965e+09	3.020e+09	100.0000
2	Tooth	184211	1.300e+00	3.766e+09	1.794e+10	100.0000

	flank					

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

15.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	9500.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	99.5455
Reliability R, gears subsystem (%)	[R_{subG}]	99.5455
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	11049.6456
Service life H, tooth flanks subsystem (h)	[H_{subH}]	9947.1741
Service life H, gears subsystem (h)	[H_{subG}]	9947.1741

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

16.2 Calculations and factors

- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with Z_β = 1/(COS(β)^0.5)
- Details of calculation method:
  c_v according to Method B
  K_v according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 668)